Roll Number: - _____

Thapar Institute of Engineering & Technology, Patiala
Department of Computer Science & Engineering
Auxiliary Examination

Course Code: UCS510 / UCS319	B. E. 3rd year (COE/CSE)
Course Name: Computer Architecture and Organization	
26/08/2025	5:30 PM - 8:30 PM
Time: 3 Hours, M. Marks: 100	Faculty: Dr. Yadwinder Singh

Note: Attempt any five (5) questions in serial order. Assume missing data if any.

Q.1.	a. Minimize the following Boolean functions using K-map and write the simplified expressions. F (A, B, C, D) = Σ(0, 1, 2, 5, 7, 8, 9, 10, 13, 15) F (A, B, C, D) = π(3, 4, 6, 11, 14) b. The two-word instruction at address 300 and 301 is a "load to AC" instruction with an address field equal to 600. The first word of the instruction specifies the operation code and mode, and the second word specifies the address part. PC has the value 300 for fetching this instruction. The content of processor register R1 is 500, and the content of an index register XR is 200. The contents of memory at locations 499, 500, 600, 700, 800, 802, 900, 902 are 550, 800, 900, 1000, 650, 350, 400, 700 respectively. Determine the Effective Address and the content of AC for the following addressing modes: (i) Direct address (ii) Indirect address (iii) Relative address (iv) Indexed address (v) Register indirect	(10+10)
Q.2.	a. Use BOOTH's Algorithm to show the multiplication of the following signed decimal **(-5 X 10)** numbers. The number of bits to represent each digit should be 5. Show each step very clearly. b. Differentiate between: (i) microprogrammed and hardwired control Unit (ii) RISC and CISC	(10+10)
Q.3.	a. A 4-way set-associative cache memory unit with a capacity of 16 KB is built using a block size of 8 words. The word length is 32 bits. The size of the physical address space is 4 GB. Determine the number of bits for TAG, Set and Word offset fields respectively. b. Consider a non-pipelined processor with a clock rate of 3.6 gigahertz and average cycles per instruction of 5. The same processor is upgraded to a pipelined processor with six stages, but due to the internal pipeline delay, the clock speed is reduced to 2.5 gigahertz. Assume there are stalls of 1 ns after each stage in the pipeline. Find the speed up achieved in this pipelined processor. Show each step clearly.	(10+10)

<table>
<tr><td>Q.4.</td><td>a. Consider a processor with memory unit of having 4096 words and each word of size 16 bit. Explain the special purpose registers namely PC, AR, DR, TR, IR, AC, INPR and OUTR along with their size for the given configuration in detail.
b. Describe different pipelining hazards and how they can impact the performance of a pipelined processor. Explain each of them with suitable example (set of instructions where the particular hazard is occurring) and how they can be resolved.</td><td>(10+10)</td></tr>
<tr><td>Q.5.</td><td>a. What is DMA? Draw a typical block diagram of a DMA controller and explain how it is used for direct data transfer between memory and peripherals.
b. Explain BNZ, BUN, and BSA and INZ Instruction using their micro-operation.</td><td>(10+10)</td></tr>
<tr><td>Q.6.</td><td>a. Represent the following in single precision IEEE 754 floating-point format. Show each step clearly.
1. $(-0000.1111)_2$
2. $(14.625)_{10}$
b. Design an arithmetic circuit with one selection variable S_0 and two 4-bit data inputs A and B. The circuit generates the following four arithmetic operations in conjunction with the input carry C_{in}. Draw the logic diagram for
<table>
<tr><td></td><td>$C_{in}=0$</td><td>$C_{in}=1$</td></tr>
<tr><td>$S_0=0$</td><td>A+B</td><td>Transfer A</td></tr>
<tr><td>$S_0=1$</td><td>A+1</td><td>A-B</td></tr>
</table></td><td>(10+10)</td></tr>
</table>

Checked Answer Sheets will be shown on 2/9/2025 (Tuesday), 11:30am, Cabin-L547

Roll Number:

Thapar Institute of Engineering & Technology, Patiala

Department of Computer Science & Engineering

B. E.(COE/CSE) : 3rd Year
Special AUX
Date: Sep, 25, 2025
Maximum Marks: 100

Course Code: UCS510
Course Name: Computer Organization and Architecture
Time: 5:30 PM - 8:30 PM
Faculty: JPK

Note: All questions carry equal marks. Attempt any 10 questions. Assume missing data if any.

Q.1	Perform the signed multiplication of $(-13)_{10}$ and 7_{10} using Booth's Algorithm. Assume 6-bit registers for both the multiplicand and multiplier. Show all intermediate steps clearly, including the final binary result and its decimal interpretation.	10
Q.2	Discuss in detail the key differences between (a) RISC (Reduced Instruction Set Computer) and CISC (Complex Instruction Set Computer) architectures. (b) Direct and indirect memory addressing modes (c) Hardwired and Micro-programmed control	10
Q.3	Convert the following numbers in $IEEE-754$ floating point $32-bit$ short precision format. **(a). $(15.25)_{10}$ (b). $(CA64)_{H}$**	10
Q.4	Consider a 16-way set associative mapped cache with block size 2 KB. The size of main memory is 32 GB and there are 12 bits in the tag. Compute: (a). Size of cache memory (b). Tag directory size (c). Number of Sets in the Cache	10
Q.5	Explain the working of a Direct Memory Access (DMA) controller in a modern computer system. Draw a detailed block diagram, identify and explain all involved signals, and describe the sequence of events during a DMA data transfer.	10
Q.6	Clearly define what is meant by hazards in instruction pipelining within computer architecture. Discuss in detail the various types of pipeline hazards that can occur during instruction execution. For each type of hazard, illustrate with a relevant example, and explain at least one effective technique or solution commonly used to resolve or minimize its impact on pipeline performance.	10
Q.7	Design an arithmetic logic unit which can be combined into one ALU with common selection variables such that they can perform the following operations for a 4-bit ALU:	10

S_2	S_1	S_0	C_{in}	**Operations**
0	0	0	0	F = A
0	0	1	1	F = A + B + 1
1	1	0	X	F = A $\oplus$B
1	1	1	X	F = A'

Q.8	Simplify the given 5-variable K-map. Show each step and grouping of minterms clearly. $F(A,B,C,D,E) = \sum m(1,2,3,5,7,11,13,17,19,23,29,31)$	10

Q.9	A computer uses a memory unit with $1024K$ words of 32 bits each. A binary instruction code is stored in one word of memory. The instruction has four parts: an indirect bit, an operation code, a register code part to specify one of 128 registers, and an address part. a) How many bits are there in the operation code, the register code part, and the address part? b) Draw the instruction word format and indicate the number of bits in each part. c) How many bits are there in the data and address inputs of the memory?	10
Q.10	Consider a non-pipelined processor with a clock rate of 2.5 gigahertz and an average cycles per instruction (CPI) of 4. The same processor is upgraded to a pipelined processor with five stages; but due to internal pipeline delay, the clock speed is reduced to 2 gigahertz. Assume that there are no stalls in the pipeline. i) Calculate the speedup achieved by the pipelined processor compared to the non-pipelined processor. ii) If you need to execute 1000 independent instructions, compute the total execution time required on both the non-pipelined and pipelined processors.	10
Q.11	Consider the following program segment. Here $R1$, $R2$, and $R3$ are general purpose registers. **Stmt** | **Symbolic** | **Explanation** MOV R1, (3000) | R1 ← M[3000] | Load mem[3000] into R1 MOV R2, (R3) | R2 ← M[R3] | Load mem[R3] into R2 ADD R2, R1 | R2 ← R2+R1 | Add R1 to R2 MOV (R3), R2 | M[R3] ← R2 | Store R2 in mem[R3] INC R3 | R3 ← R3+1 | Increment R3 DEC R1 | R1 ← R1− | Decrement R1 IF R1 $\neq$ 0 GOTO LOOP | if R1 $\neq$ 0, PC←LOOP | Jump to LOOP if R1$\neq$0 HALT | − | Stop execution Assume that the content of memory location 3000 is 10, and the content of register $R3$ is 2000. The content of each memory location from 2000 to 2010 is 100. The program is loaded starting at memory location 1000. All numbers are in decimal. The memory is word addressable. i) Determine the number of memory references made for accessing data during the complete execution of the program. ii) What is the final content of memory location 2010 after execution?	10